



# KunHo Heo

✉ hkh7710@gmail.com | 🏠 heokunho.github.io | 🎓 Google Scholar

## Research Interests

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### 3D Vision: Scene Understanding

- Scene Graph, Affordance Grounding

### 3D Vision: Character Animation

- Human Motion Generation, Human-Object Interaction

### 2D Vision: Perception

- Few-Shot Object Detection

## Education

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### Kyung Hee University

Mar. 2025 – Feb. 2027 (Expected)

M.S. in Software Convergence

- GPA: 4.45 / 4.5

### Kyung Hee University

Mar. 2021 – Feb. 2025

B.S. in Software Convergence

- GPA: 4.04 / 4.5

## Research Experience

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### Visual Science Lab [🔗](#), Kyung Hee University

Mar. 2025 – Present

M.S. Student (**Lab Manager**)

- Advisor: Prof. MyeongAh Cho

### Visual Science Lab [🔗](#), Kyung Hee University

Sep. 2024 – Feb. 2025

Undergraduate Research Intern

- Advisor: Prof. MyeongAh Cho

### \*IIIXR Lab [🔗](#), Kyung Hee University *\*relocated to Korea University*

Jan. 2023 – Jun. 2024

Undergraduate Research Intern

- Advisor: Prof. HyeongYeop Kang

## Publications

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### International Conference

#### Rethinking Prototype-based Similarity Learning for Few-Shot Object Detection

**KunHo Heo**<sup>\*</sup>, Seungjae Kim<sup>\*</sup>, Wongyu Lee<sup>\*</sup>, SuYeon Kim, MyeongAh Cho *\*Equal Contribution*

Under Review

#### ParTY: Part-Guidance for Expressive Text-to-Motion Synthesis

Paper [🔗](#)

**KunHo Heo**, SuYeon Kim, Yonghyun Gwon, Youngbin Kim, MyeongAh Cho

IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR 2026**)

## Object-Centric Representation Learning for Enhanced 3D Semantic Scene Graph Prediction

Paper [↗](#)

KunHo Heo\*, GiHyun Kim\*, SuYeon Kim, MyeongAh Cho \*Equal Contribution

The Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS 2025)

## International Journal

### Revisiting Object-Centric Learning for Long-Tailed 3D Semantic Scene Graph Prediction

GiHyun Kim, KunHo Heo, SuYeon Kim, MyeongAh Cho

Under Review (IEEE Transactions on Image Processing)

## Domestic Conference

### Word-Motion Cross Attention for Fine-Grained Text Guided Human-Human Interaction Motion Generation

Yonghyun Gwon, KunHo Heo, Youngkwan Cho, SuYeon Kim, Hayoung Lee, MyeongAh Cho

Korea Software Congress (KSC), 2025

### Distance-based Bi-directional Edge Update Graph Neural Network for 3D Scene Graph Generation

SuYeon Kim, GiHyun Kim, KunHo Heo, MyeongAh Cho

Korea Computer Congress (KCC), 2025

### Enhancing Text-driven Human Motion Generation via Part Transfer and Sequential Attention Learning

KunHo Heo, Seung Joon Lee, Hayoung Lee, MyeongAh Cho

Korea Software Congress (KSC), 2024

### Body-Part Embedding for Text-driven Human Motion Generation

KunHo Heo, HyeongYeop Kang

Korea Computer Congress (KCC), 2024

### Personality labeling methods with motion parameters

KunHo Heo, HyeongYeop Kang

Korea Software Congress (KSC), 2023

## Projects

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### Development of AGI platform for multisensory social robot

Institute for Information communication  
Technology Planning and Evaluation (IITP)  
Apr. 2025 - Dec. 2025

- **Project Manager**
- Developed the **Vision system** for multisensory perception
- Researched **Few-Shot Object Detection** to enhance adaptability to dynamic environments under limited data

### Development of AI technology capable of robustly recognizing abnormal and dangerous situations and behaviors during night and bad weather conditions

Korea Planning & Evaluation Institute  
of Industrial Technology (KEIT)  
Jul. 2024 - Mar. 2028

- **Image restoration** and **Data pre-processing** under adverse visual conditions

### Anomaly pre-detection and response for intelligent surveillance system: high-level interpretation and response based on the visual-language fusion

National Research Foundation  
of Korea (NRF)  
Sep. 2024 - Aug. 2027

- **Vision-language model integration** for surveillance anomaly interpretation and response

## Patents

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### Domestic Patent

**Method and Apparatus for Generating Whole-Body Motion by Synthesizing Part-wise Motions** Dec. 2025

- KR-Application No. 10-2025-0200238

**Method and Apparatus for Generating 3D Scene Graphs** Dec. 2025

- KR-Application No. 10-2025-0188361

### Software Copyright

**Improved Sound Source Localization Model via Audio Signal Enhancement** Sep. 2025

- KR-Registration No. C-2025-035472

## Honors & Awards

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**Outstanding Researcher Scholarship** 2025, 2026

- Kyung Hee University

**Academic Scholarship (Awarded in 7 of 8 Semesters)** 2021 – 2024

- Kyung Hee University

**Grand Prize** Dec. 2024

- Undergraduate Paper Competition, Korea Software Congress (**KSC**)

**Grand Prize (Overall 1st Prize across all categories)** Nov. 2024

- SW Festival, Kyung Hee University

**1st Prize in Immersive Media Category** Nov. 2024

- SW Festival, Kyung Hee University

**National Research Foundation (NRF) President's Award** Nov. 2024

- Immersive Media Competition, Convergence and Open Sharing System (**COSS**)

**Grand Prize** May. 2024

- AI&SW Hackathon, Kyung Hee University

**1st Prize in Immersive Media Category** Nov. 2023

- SW Festival, Kyung Hee University

## Teaching Experience

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**Teaching Assistant** Spring 2025, Spring 2026

- Artificial Intelligence and Game Programming

**Teaching Assistant** Fall 2025

- Game Engineering
- Open Source Software Development Methods and Tools
- Design Thinking